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Cooling mold, and apparatus and method for manufacturing resin molded article

Abstract

A cooling mold for cooling a resin preform that has a gate portion protruding outward from a center of a bottom portion, in which an accommodating space that accommodates the preform is formed in the cooling mold, a bottom region facing the bottom portion of the preform in the accommodating space has a shape following the bottom portion of the preform, and an air sucking hole for sucking air is formed in the bottom region at a position shifted from the center of the bottom portion of the preform. In the accommodating space, a region facing a body portion of the perform and a bottom region facing the bottom portion of the perform are formed as a single member.

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a cooling mold, and an apparatus and a method for manufacturing a resin molded article.
Description of the Related Art
(2) In blow molding of a resin container, a configuration has been proposed in which a resin preform released from an injection mold is forcibly cooled using a cooling mold different from the injection mold (for example, see WO 2017-073791 A).
(3) Regarding this type of cooling mold, there is also known a mold that sucks air between a mold and a preform to bring the preform into close contact with a surface of the mold (for example, see JP 2004-521779 A and JP 2509803 B2).
(4) In a bottom portion of an injection-molded preform, a gate portion that becomes a resin introduction mark from a hot runner into an injection mold is formed in a manufacturing process. When blow-molding a container from a preform, it is preferred, in terms of appearance and quality of a container after molding, to completely eliminate a gate portion of a preform from a bottom portion thereof. However, it is actually difficult to mechanically cut off a gate portion of a preform with high accuracy due to restrictions such as molding conditions of a preform and the like and apparatus configurations.
(5) On the other hand, in a blow molding manufacturing cycle, it is desired to further shorten molding cycle time of a container. For example, when a process of mechanically cutting off a gate portion of a preform is added, molding cycle time of a container is lengthened by time of the added cutting process.
SUMMARY OF THE INVENTION
(6) One aspect of the present invention is a cooling mold for cooling a resin preform that has a gate portion protruding outward from a center of a bottom portion, in which an accommodating space that accommodates the preform is formed in the cooling mold, a bottom region facing the bottom portion of the preform in the accommodating space has a shape following the bottom portion of the preform, and an air sucking hole that sucks air is formed in the bottom region at a position shifted from the center of the bottom portion of the preform.

(7) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a plan view schematically illustrating a configuration of a blow molding apparatus.
(2) FIG. **2** is a view schematically illustrating conveyance of a preform in an injection molding unit and a cooling unit.
(3) FIGS. **3**A to **3**C are views illustrating an example of a preform.
(4) FIG. **4** is a front view illustrating a configuration example of a cooling unit.
(5) FIGS. **5**A and **5**B are views illustrating a configuration example of a cooling pot.
(6) FIGS. **6**A and **6**B are views illustrating removal of a gate portion by a cooling pot.
(7) FIG. **7** is a graph illustrating an example of temperature changes of a preform in a blow molding method according to the present embodiment.
(8) FIG. **8** is a diagram schematically illustrating a configuration of an injection molding apparatus.
BRIEF DESCRIPTION OF THE DRAWINGS
(9) Hereinafter, embodiments of the present invention will be described with reference to the drawings.
(10) In the embodiments, for easy understanding, structures and elements other than a main part of the present invention will be described in a simplified or an omitted manner. In the drawings, identical elements are denoted by identical reference numerals. Note that shapes, dimensions, and the like of respective elements illustrated in the drawings are schematically illustrated, and do not indicate actual shapes, dimensions, and the like.
(11) <Description of Blow Molding Apparatus>
(12) First, a blow molding apparatus **100**, an example of a manufacturing apparatus for manufacturing a resin container, will be described with reference to FIG. **1**. FIG. **1** is a plan view schematically illustrating a configuration of a blow molding apparatus. FIG. **2** is a view schematically illustrating conveyance of a preform in an injection molding unit and a cooling unit.
(13) The blow molding apparatus **100** according to the present embodiment performs a blow molding method called a 1.5 stage method having advantages of both a hot parison method and a cold parison method. In the blow molding method in the 1.5 stage method, basically similarly to the hot parison method (one stage method), a preform having heat from injection molding is blow-molded to manufacture a container. However, a cycle of blow molding in the 1.5 stage method is set to be shorter than a cycle of injection molding of a preform. A plurality of preforms molded in one injection molding cycle is blow-molded in a plurality of blow molding cycles. Although not particularly limited, a ratio (N:M) of the number (N) of preforms simultaneously injection-molded and the number (M) of containers simultaneously blow-molded is set to 3:1, for example.
(14) As illustrated in FIG. **1**, the blow molding apparatus **100** includes an injection molding unit **110**, a cooling unit **120**, a heating unit **130**, and a blow molding unit **140**.
(15) The blow molding apparatus **100** also includes a continuous conveying unit **150** that conveys preforms **200** carried out from the cooling unit **120** to the blow molding unit **140** via the heating unit **130**. The continuous conveying unit **150** is a conveying device that continuously conveys conveying jigs **152** holding the preforms **200** along a loop-shaped conveying line **151** that has a plurality of curved portions. In other words, the continuous conveying unit **150** can repeatedly convey the conveying jigs **152** each along the loop-shaped conveying line **151**.
(16) The injection molding unit **110** injection-molds the bottomed tubular preforms **200** that are resin molded articles.
(17) As illustrated in FIG. **2**, the injection molding unit **110** includes core molds **111** disposed

above, cavity molds **112** disposed below, and a mold clamping mechanism **114** that clamps the core molds **111** and the cavity molds **112** with tie bars **113**. The injection molding unit **110** injection-molds the preforms **200** by filling injection spaces formed by the core molds **111** and the cavity molds **112** with a resin material (raw material) from an injection device (not illustrated).
(18) The injection molding unit **110** according to the present embodiment simultaneously molds, for example, 3 rows×4 (N=12) preforms **200**. In addition, the preforms **200** are molded in an upright state with neck portions facing upward in the injection molding unit **110**, and the preforms **200** are conveyed in the upright state in the injection molding unit **110**.
(19) As illustrated in FIG. **2**, the injection molding unit **110** includes a receiving unit **115** that takes out injection-molded preforms to the outside of the injection molding unit **110**.
(20) The receiving unit **115** can move in a horizontal direction (X direction in the drawing) from a receiving position on a lower side of the core molds **111** to a delivery position outside a space surrounded by the tie bars **113**.
(21) The receiving unit **115** holds 12 cooling pots **300** that respectively accommodate 3 rows×4 preforms molded in the injection molding unit **110**.
(22) Each cooling pot **300** is an example of a cooling mold, and has an accommodating space for each preform **200**, corresponding to an outer shape of the preform **200**. The cooling pot **300** of the receiving unit **115** has a function of cooling the preform by contacting the accommodated preform **200**. The cooling pot **300** also has a function of removing a gate portion of the preform **200**. A configuration of the cooling pot **300** will be described later.
(23) Further, the receiving unit **115** includes a mechanism (not illustrated) that adjusts an interval (an interval in the X direction in the drawing) between rows of the cooling pots **300** while moving from the receiving position to the delivery position. As a result, the receiving unit **115** converts the interval between the rows of the preforms **200** from a wide pitch state of the receiving position to a narrow pitch state of the delivery position.
(24) Here, an example of the preform **200** applied in the present embodiment will be described with reference to FIGS. **3**A to **3**C. FIG. **3**A is a longitudinal sectional view of the preform **200** in the upright state as viewed from a front direction, and FIG. **3**B is a plan view of the preform **200**. FIG. **3**C is a longitudinal sectional view illustrating the preform **200** in a state where the gate portion is removed by the injection molding unit **110** or the cooling unit **120** according to the present embodiment.
(25) An entire shape of the preform **200** is a bottomed cylindrical shape in which one end side is opened and the other end side is closed. The preform **200** includes a body portion **201** formed in a cylindrical shape, a bottom portion **202** that closes the other end side of the body portion **201**, a gate portion **203** formed in the bottom portion **202**, and a neck portion **204** formed in an opening on one end side of the body portion **201**.
(26) The gate portion **203** is a resin introduction mark from a hot runner, and is formed so as to protrude outside the bottom portion **202** at a center of the bottom portion **202**.
(27) A raw material of the preform **200** is a thermoplastic synthetic resin, and can be appropriately selected according to uses of containers. Specific types of materials include polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polycyclohexanedimethylene terephthalate (PCTA), Tritan (registered trademark: co-polyester manufactured by Eastman Chemical Co., Ltd.), polypropylene (PP), polyethylene (PE), polycarbonate (PC), polyethersulfone (PES), polyphenylsulfone (PPSU), polystyrene (PS), cyclic olefin polymer (COP/COC), polymethylmethacrylate (PMMA: acrylic), polylactic acid (PLA), and the like.
(28) The preform **200** injection-molded by the injection molding unit **110** is supplied from the injection molding unit **110** to the cooling unit **120**. The cooling unit **120** forcibly cools the preform **200** molded by the injection molding unit **110**. The preform **200** is carried out from the cooling unit **120** in a state of being cooled to a predetermined temperature, and is continuously conveyed along the conveying line **151**.

(29) As illustrated in FIG. **2**, a conveying device **180** that conveys the preform **200** in the upright state from the receiving unit **115** to the cooling unit is provided between the injection molding unit **110** and the cooling unit **120**. The conveying device **180** includes a holding unit **181** that holds the neck portion of the preform **200** in the upright state, and can move the holding unit **181** in a vertical direction (Z direction in the drawing) and the horizontal direction (X direction in the drawing) by an air cylinder (not illustrated).
(30) As illustrated in FIG. **4**, the cooling unit **120** includes an inverting unit **121**. The inverting unit **121** is invertible around a shaft **122** extending in the X direction in the drawing as a rotation axis, and is movable up and down in the Z direction (vertical direction) in the drawing. On a first surface **121***a* illustrated on an upper side in the drawing of the inverting unit **121** and a second surface **121***b* facing the first surface **121***a*, 12 cooling pots **300** are disposed on each of the surfaces in order to accommodate 3 rows×4 preforms **200**.
(31) In the present embodiment, the cooling pots **300** disposed on the first surface **121***a* and the second surface **121***b* in the inverting unit **121** have a configuration similar to the configuration of the cooling pots **300** of the receiving unit **115**. The cooling pots **300** in the inverting unit **121** are cooled by a refrigerant circulating through a refrigerant passage (not illustrated) provided in the inverting unit **121**. The cooling pots **300** each in the inverting unit **121** have a function of sucking and holding the accommodated preform **200** and a function of removing the gate portion of the preform.
(32) In the following description, the cooling pots **300** disposed on the first surface **121***a* in the inverting unit **121** are also referred to as first cooling pots **300***a*, and the cooling pots **300** disposed on the second surface **121***b* in the inverting unit **121** are also referred to as second cooling pots **300***b*.
(33) In addition, the inverting unit **121** reverses the preforms **200** in the upright state received from the conveying device **180** to an inverted state in which the neck portions face downward during cooldown time. Then, the preforms **200** in the inverted state are delivered to the conveying jigs **152** of the continuous conveying unit **150**, disposed in a plurality of rows below the cooling unit **120**. The conveying jigs **152** holding the preforms **200** are sequentially conveyed along the conveying line **151** by driving forces of sprockets **154** or the like.
(34) The heating unit **130** heats the preforms **200** in the inverted state continuously conveyed by the continuous conveying unit **150** to an appropriate stretching temperature. The heating unit **130** includes a plurality of heaters (not illustrated) disposed at predetermined intervals along the conveying line **151** on both sides of the conveying line **151**. In the heating unit **130**, the preforms **200** in the inverted state are heated while rotating about axial directions of the preforms **200**, and the entire preforms **200** are uniformly heated.
(35) Further, the blow molding apparatus **100** includes an intermittent conveying unit **160** and a delivery unit **170** on a downstream side of the heating unit **130** in the conveying line **151**.
(36) The intermittent conveying unit **160** holds a plurality of (M, e.g., four) preforms **200** heated by the heating unit **130** and intermittently conveys the preforms **200** to the blow molding unit **140**. The delivery unit **170** delivers the preforms **200** continuously conveyed by the continuous conveying unit **150** from the conveying line **151** to the intermittent conveying unit **160**.
(37) In the present embodiment, a plurality of (for example, eight) conveying jigs **152** continuous in a conveying direction is connected by a connecting member (not illustrated). Then, the continuous conveying unit **150** repeats driving and stopping by sprockets **154***a* on the conveying line **151** on a downstream side of a curved conveying unit **155** curved at a predetermined radius, thereby supplying the (M, e.g., four) preforms **200** to the delivery unit **170** at a time.
(38) The delivery unit **170** includes a reversing device (not illustrated) at a delivery position P**0**. The preforms **200** conveyed in the inverted state along the conveying line **151** are inverted by the reversing device disposed on the upper side of the preforms **200** at the delivery position P**0** to be in the upright state. In addition, the delivery unit **170** includes, for example, a lifting device (not

illustrated) that lifts and lowers the reversing device, and delivers the preforms **200** in the upright state to the intermittent conveying unit **160** in a state of being lifted to a predetermined position (delivery position P**1**).
(39) The intermittent conveying unit **160** grips the neck portion of each of the preforms **200** in the upright state by an openable and closable blow conveying chuck member (not illustrated) provided in the intermittent conveying unit **160**. Then, the chuck member (not illustrated) of the intermittent conveying unit **160** grips the neck portion of the preform **200** at the delivery position P**1** located above the delivery position P**0**, and moves the preform **200** from the delivery position P**1** to a blow molding position P**2**. As a result, the preforms **200** are conveyed to the blow molding unit **140** at predetermined intervals.
(40) The blow molding unit **140** includes a pair of blow cavity molds **141** that are split molds corresponding to shapes of containers and an air introduction member (not illustrated) that also serves as a stretching rod. In the blow molding unit **140**, predetermined number of the preforms **200** received from the delivery unit **170** are conveyed to the blow cavity molds **141**, and the preforms **200** are subjected to stretch blow molding by the blow cavity molds **141** to manufacture containers.
(41) The containers manufactured by the blow molding unit **140** are conveyed to a taking-out position P**3** outside the blow molding unit **140** by the intermittent conveying unit **160**.
(42) <Description of Cooling Pots>
(43) Next, configuration examples of the cooling pots **300** of the injection molding unit **110** and the cooling unit **120** will be described with reference to FIGS. **5** and **6**.
(44) As described above, the first cooling pots **300***a* and the second cooling pots **300***b* of the cooling unit **120** are similar to the cooling pots **300** of the receiving unit **115**. Therefore, here, the configuration of the cooling pots **300** of the receiving unit **115** will be described, and redundant description will be omitted.
(45) FIG. **5**A is a plan view of each of the cooling pots **300**, and FIG. **5**B is a sectional view taken along a line Vb-Vb in FIG. **5**A.
(46) The cooling pot **300** has a bottomed cylindrical shape as a whole, and is a cooling mold into which the preform **200** can be inserted from an upper surface side. The cooling pot **300** has an accommodating space **301** capable of accommodating the body portion **201** and the bottom portion **202** of the preform **200** and having an open upper surface side. An internal shape of the accommodating space **301** is a shape following outer shapes of the body portion **201** and the bottom portion **202** of the preform **200**.
(47) In the accommodating space **301** of the cooling pot **300**, air sucking holes **302** for sucking air between the mold and the preform **200** are formed in a bottom region **301***a* facing the bottom portion **202** of the preform **200**. Each of the air sucking holes **302** is connected to an air sucking pump (not illustrated) via an air flow path **303** formed in the cooling pot **300**.
(48) The air sucking hole **302** is formed at a position shifted from a center of the bottom portion of the preform **200** in the bottom region **301***a* of the accommodating space **301**. In other words, there is no air sucking hole **302** at a position facing the gate portion **203** of the preform **200** in the cooling pot **300** (center of an axial direction of the cooling pot **300**), and a surface of the bottom region **301***a* following the outer shape of the bottom portion **202** of the preform **200** faces the gate portion **203**.
(49) The air sucking holes **302** are formed in the bottom region **301***a* of the accommodating space **301** so as to be rotationally symmetric with respect to the center of the bottom portion of the preform **200**. FIG. **5**A illustrates an example in which four of the air sucking holes **302** are disposed at intervals of 90 degrees in a circumferential direction of an inner peripheral surface of the bottom region **301***a* so as to be point-symmetric with respect to the center of the bottom region **301***a* (position faced by the center of the bottom portion of preform **200**, the center of the axial direction of the cooling pot **300**). Note that the number of the air sucking holes **302** provided may be a

number other than four (two, three, or more integers) as long as disposition is rotationally symmetric with respect to the center of the bottom portion of the preform **200**. Further, an annular air sucking hole (not illustrated) concentric with the center of the bottom portion of the preform **200** may be formed.
(50) FIG. **6**A is a schematic view illustrating a state before air between the cooling pot **300** and the preform **200** is sucked, and FIG. **6**B is a schematic view illustrating a state after air between the cooling pot **300** and the preform **200** is sucked.
(51) When air is sucked from the air sucking holes **302** in a state where the preform **200** is disposed in the accommodating space **301** of the cooling pot **300** (see FIG. **6**A), the preform **200** is drawn inward into the accommodating space **301** and comes into close contact with the cooling pot **300** (see FIG. **6**B).
(52) As a result, the body portion **201** and the bottom portion **202** of the preform **200** come into surface contact with a surface of the accommodating space **301**, and the preform **200** is efficiently cooled by heat exchange with the cooling pot **300**. Since a shape of the accommodating space **301** of the cooling pot **300** follows the outer shape of the preform **200**, the shape of the preform **200** is maintained by the cooling pot **300** at a time of cooling.
(53) Here, since the preform **200** has residual heat from injection molding, the gate portion **203** is easily deformed. Therefore, when the preform **200** is drawn inward into the accommodating space **301** as described above, the gate portion **203** located at the center of the bottom portion of the preform **200** is collapsed on the surface of the opposing bottom region **301***a*. As a result, as illustrated in FIG. **6**B, the shape of the bottom portion of the preform becomes a curved surface following the bottom region **301***a* of the accommodating space **301**.
(54) As described above, in the present embodiment, the gate portion **203** is removed from the bottom portion **202** of the preform **200** with high accuracy in a process of cooling the preform **200** with the cooling pot **300**.
(55) Further, the air sucking holes **302** are located at the positions shifted from the center of the bottom portion of the preform **200**, but are disposed so as to be rotationally symmetric. Therefore, sucking power when the preform **200** is drawn substantially uniformly acts on the bottom portion **202** of the preform **200**. Thus, it is possible to suppress distortion from occurring in the bottom portion **202** of the preform **200** when the preform **200** is drawn inward into the accommodating space **301**.
(56) The first cooling pot **300***a* and the second cooling pot **300***b* of the cooling unit **120** also function similarly to the cooling pot **300** described above. Although the cooling unit **120** inverts the preform **200** received in the upright state to the inverted state, the cooling unit **120** can adsorb and hold the preform **200** in the inverted state by sucking air from the air sucking holes **302**.
(57) <Description of Blow Molding Methods>
(58) Next, a blow molding method by the blow molding apparatus **100** according to the present embodiment will be described.
(59) FIG. **7** is a diagram for explaining temperature changes of the preform **200** in the blow-molding method according to the present embodiment. In FIG. **7**, a vertical axis represents a temperature of the preform **200**, and a horizontal axis represents time. In FIG. **7**, examples of temperature changes of the preform according to the present embodiment are indicated by (A) in FIG. **7**. Examples of temperature changes of the preform in a comparative example (conventional method) described later are indicated by (B) in FIG. **7**.
(60) (1) First, in the injection molding unit **110**, a resin is injected from an injection device into preform-shaped mold spaces formed by the core molds **111** and the cavity molds **112** to manufacture N preforms **200**.
(61) In the present embodiment, the injection molding unit **110** is opened immediately after completion of resin filling or after minimum cooldown time provided after resin filling, and the preforms **200** are released from the core molds **111** and the cavity molds **112** in a high temperature

state in which the outer shapes of the preforms **200** can be maintained. In short, in the present embodiment, when the resin material is injected at a temperature equal to or higher than a melting point of the resin material, only minimum cooling of the preforms **200** after injection molding is performed in the injection molding unit **110**, and the preforms **200** are cooled in the cooling pots **300** of the receiving unit **115** or the first cooling pots **300***a* or the second cooling pots **300***b* in the cooling unit **120**.
(62) In the present embodiment, time (cooldown time) for cooling the resin material by the injection molding unit **110** after completion of injection of the resin material is preferably ½ or less with respect to time (injection time) for injecting the resin material. The time for cooling the resin material can be made shorter than the time for injecting the resin material depending on weight of the resin material. The time for cooling the resin material is more preferably ⅖ or less, still more preferably ¼ or less, and particularly preferably ⅕ or less with respect to the time for injecting the resin material. Since the cooldown time is significantly shortened as compared with the comparative example described later, a skin layer (surface layer in a solidified state) of a preform is formed to be thinner than before, and a core layer (inner layer in a softened or molten state) is formed to be thicker than before. In other words, as compared with the comparative example, a preform having a large heat gradient between the skin layer and the core layer and having high residual heat at a high temperature is molded.
(63) On the other hand, as a comparative example, when the preforms **200** are cooled in the core molds **111** and the cavity molds **112**, examples of temperature changes of the preforms ((B) in FIG. **7**) will be described.
(64) In the comparative example, each of the preforms **200** is cooled to a temperature lower than one in the present embodiment in the molds of the injection molding unit **110**. Therefore, in the comparative example, the molding cycle time of a preform becomes longer than that in the present embodiment, and as a result, the molding cycle time of a container also becomes longer.
(65) (2) The N preforms **200** manufactured in the injection molding unit **110** are delivered in the upright state to the cooling pots **300** of the receiving unit **115** and carried out from the injection molding unit **110**. In each of the cooling pots **300**, each of the preforms **200** is cooled and the gate portion **203** is removed as described above.
(66) (3) At the delivery position of the receiving unit **115**, the N preforms **200** are delivered to the conveying device **180** and conveyed in the upright state to the cooling unit **120**. In the cooling unit **120**, the conveyed N preforms **200** are accommodated in either the first cooling pots **300***a* or the second cooling pots **300***b*. Thereafter, the cooling unit **120** is reversed, and the preforms **200** are in the inverted state and delivered to the conveying jigs **152** of the continuous conveying units **150**.
(67) In the first cooling pots **300***a* and the second cooling pots **300***b* in the cooling unit **120**, the preforms **200** are cooled. At this time, in the first cooling pots **300***a* and the second cooling pots **300***b*, the preforms **200** are sucked and held by air suction, and the gate portions **203** are further removed. The gate portions **203** are removed twice, that is, in the cooling pots **300** of the receiving unit **115** and the first cooling pots **300***a* or the second cooling pots **300***b* of the cooling unit **120**, so that the gate portions **203** of the preforms **200** can be removed with higher accuracy.
(68) Furthermore, in the present embodiment, for example, the N preforms **200** in the upright state, injection-molded in the m-th cycle are held by the first cooling pots **300***a*, then inverted by the inverting unit **121**, and cooled in the inverted state. During this period, the N preforms **200** in the upright state, injection-molded in a cycle following the m-th cycle (m+1-th cycle) are held and cooled in the second cooling pots **300***b*. In short, the cooling unit **120** can simultaneously cool the preforms **200** in different cycles on the first surface **121***a* and the second surface **121***b*.
(69) As described above, the cooling unit **120** can forcibly cool the preforms **200** for a time equal to or longer than the cycle time of injection molding when the injection molding unit **110** injection-molds the N preforms **200**.
(70) Since the preforms **200** forcibly cooled in the cooling unit **120** do not need to be cooled to a

room temperature and the preforms **200** have heat from injection molding, it is possible in the present embodiment as well to share advantages of energy efficiency in the one stage method.
(71) Although the N preforms **200** immediately before heating have residual heat from injection molding, clear temperature differences may occur among the cycles of blow molding described later, depending on natural cooling time due to time differences among the cycles of blow molding. The forced cooling in the cooling unit **120** is effective, when the N preforms **200** simultaneously injection-molded are heated at different heating start timings, in suppressing temperature differences of the preforms **200** immediately before heating in each cycle of blow molding.
(72) (4) The preforms **200** in the inverted state are continuously conveyed along the conveying line **151** of the continuous conveying unit **150** and pass through the heating unit **130**. As a result, the preforms **200** are subjected to temperature equalization and removal of uneven temperature, and heated to an appropriate stretching temperature.
(73) Here, the N preforms **200** simultaneously injection-molded in the m-th cycle are blow-molded a plurality of times (three times) by M preforms **200**. Therefore, a line indicating temperature changes in FIG. **7** is different for each blow molding cycle. In other words, in FIG. **7**, the temperature changes of the preforms until carried into the continuous conveying unit **150** are common regardless of the blow molding cycle. However, the temperature changes of the preforms from heating to blow molding are indicated by three lines having time differences according to the cycles of blow molding.
(74) (5) The preforms **200** that have passed through the heating unit **130** are delivered to the intermittent conveying unit **160** by the delivery unit **170** for each number (M) of containers that are to be simultaneously blow-molded. The intermittent conveying unit **160** intermittently conveys the M preforms to the blow molding unit **140**.
(75) (6) In the blow molding unit **140**, the preforms **200** are subjected to stretch blow-molding, and containers are manufactured. Thereafter, the containers manufactured by the blow molding unit **140** are carried out from the blow molding unit **140** by the intermittent conveying unit **160**.
(76) Thus, a series of processes of the blow molding method is completed.
(77) Hereinafter, effects of the present embodiment will be described.
(78) According to the present embodiment, air between the accommodating spaces **301** of the cooling pots **300** for cooling the preforms **200** and the preforms **200** is sucked, and the gate portions **203** of the preforms **200** are collapsed on the surface of the bottom regions **301***a* of the accommodating spaces **301**. As a result, since shapes of the bottom portions **202** of the preforms **200** become curved surfaces following the bottom regions **301***a* of the accommodating spaces **301**, the gate portions **203** can be removed from the bottom portions **202** of the preforms **200** with high accuracy, and an aesthetic appearance of containers to be manufactured can be improved.
(79) In addition, the removal of the gate portions **203** is carried out simultaneously when the preforms **200** are accommodated in the cooling pots **300**, using the process of cooling the injection-molded preforms **200**. Therefore, in the present embodiment, since it is not necessary to newly add a process of cutting off the gate portions **203** of the preforms **200**, molding cycle time of containers is not extended by the removal of the gate portions **203**.
(80) Further, in the removal of the gate portions **203** described above, the gate portions **203** are collapsed and integrated with the bottom portions **202** of the preforms, so that fragments are not produced with the removal of the gate portions **203**. Therefore, discarding or regenerating the fragments of the gate portions **203** is not necessary, thereby reducing costs associated with manufacturing containers.
(81) Furthermore, the removal of the gate portions **203** can be performed twice, that is, in the cooling pots **300** of the receiving unit **115** and the first cooling pots **300***a* or the second cooling pots **300***b* in the cooling unit **120**. As described above, by removing the gate portions **203** twice, the gate portions **203** can be removed from the bottom portions **202** of the preforms **200** with higher accuracy.

(82) The present invention is not limited to the above embodiment, and various improvements and design changes may be made without departing from the gist of the present invention.
(83) For example, in the above embodiment, the example in which the cooling pots **300** of the present invention are disposed in the receiving unit **115** and the cooling unit **120** has been described, but the cooling pots of the present invention may be disposed in only either one thereof.
(84) In the above embodiment, as an example, the configuration in which the cooling pots **300** are applied to the blow molding apparatus in the 1.5 step method has been described. However, the cooling pots **300** according to the present embodiment may be applied to an injection molding machine without a blow molding unit to remove the gate portions.
(85) FIG. **8** is a diagram schematically illustrating a configuration of an injection molding apparatus **400**. The injection molding apparatus **400** in FIG. **8** is a apparatus used for manufacturing the preforms **200** at a high speed. The injection molding apparatus **400** includes an injection molding unit **401**, a post-cooling unit **402**, a taking-out unit **403**, and a rotating plate **405** as a conveying mechanism.
(86) The injection molding unit **401**, the post-cooling unit **402**, and the taking-out unit **403** are disposed at positions rotated by a predetermined angle (for example, 120 degrees) in a circumferential direction of the rotating plate **404**. In the injection molding apparatus **400**, rotation of the rotating plate **405** conveys the preforms **200** having the neck portions held by the rotating plate **405** to the injection molding unit **401**, the post-cooling unit **402**, and the taking-out unit **403** in this order.
(87) The injection molding unit **401** includes injection cavity molds and injection core molds (both not illustrated), and manufactures the preforms **200** by injection molding. An injection device **404** that supplies a resin material, which is a raw material of the preforms **200**, is connected to the injection molding unit **401**.
(88) The post-cooling unit **402** can provide cooling in a short time to such an extent that the preforms **200** can be discharged in a cured state by the taking-out unit **403**. The post-cooling unit **402** includes the above-described cooling pots **300**, accommodates the preforms **200** in the cooling pots **300** by suction of air, and simultaneously performs cooling and removal of the gates.
(89) The removing unit **403** releases the neck portions of the preforms **200** from the rotating plate **405** and takes out the preforms **200** to outside of the injection molding apparatus **400**.
(90) In the injection molding apparatus **400**, the post-cooling unit **402** is provided on a downstream side of the injection molding unit **401**, so that the post-cooling unit **402** can additionally cool the preforms **200**. Additional cooling of the preforms **200** in the post-cooling unit **402** can release the preforms **200** even in a high temperature state in the injection molding unit **401**, and significantly shorten the cooldown time of the preforms **200** in the injection molding unit **401**. As a result, since molding of the subsequent preforms **200** can be started early, molding cycle time of the preforms **200** in the injection molding apparatus **400** can be shortened.
(91) Furthermore, in the post-cooling unit **402** of the injection molding apparatus **400**, the gate portions can be satisfactorily removed from the preforms **200** by using the cooling pots **300** described above.
(92) In the above embodiment, the example in which the air sucking holes **302** of each of the cooling pots **300** are disposed rotationally symmetrically has been described, but the disposition of the air sucking holes **302** is not limited to the above. For example, in the two or more air sucking holes **302**, radial distances of the air sucking holes **302** from a central axis may be different. For example, in the bottom surface region **301***a* of the cooling pot **300**, the two or more air sucking holes **302** may be provided on the inner peripheral surface in a predetermined range extending in a vertical direction away from a center (the central axis of the cooling pot **300**) which the gate portion **203** of each of the preforms **200** abuts.
(93) Additionally, the embodiments disclosed herein are to be considered as illustrative in all respects and not restrictive. The scope of the present invention is indicated not by the above

description but by the claims, and it is intended that meanings equivalent to the claims and all modifications within the scope are included.

Claims

1. A mold for cooling a resin preform that has a gate portion protruding outward from a center of a bottom portion, wherein an accommodating space that accommodates the preform is formed in the mold, a bottom region facing the bottom portion of the preform in the accommodating space has an internal shape following an outer shape of the bottom portion of the preform, and a plurality of air sucking holes that sucks air is formed in the bottom region at a position shifted from the center of the bottom portion of the preform, such that there is no air sucking hole at a position facing the gate portion of the preform in the mold, and an internal surface of the bottom region following the outer shape of the bottom portion of the preform faces the gate portion, such that the mold is adapted for: sucking air from the air sucking holes of the mold and drawing the preform into the accommodating space; and collapsing the gate portion in the bottom region to remove the gate portion.
2. The mold according to claim 1, wherein, wherein the accommodating space comprises a region facing the body portion of the preform and the bottom region facing the bottom portion of the preform, wherein the region and the bottom region of the accommodating space are formed as single member.
3. The mold according to claim 1, wherein the plurality of the air sucking holes is formed in the bottom region so as to be rotationally symmetric with respect to the center of the bottom portion of the preform.
4. An apparatus for manufacturing a resin molded article, comprising: an injection molding unit configured to injection-mold a resin preform having a gate portion protruding outward from a center of a bottom portion; and a mold according to claim 1, the mold is configured for accommodating and cooling the injection-molded resin preform in the accommodating space, wherein the mold is configured for sucking air from the air sucking holes to draw the preform into the accommodating space and collapsing the gate portion in the bottom region to remove the gate portion.
5. The apparatus for manufacturing the resin molded article according to claim 4, wherein the mold is disposed in at least either a receiving unit configured to carry out the preform from the injection molding unit or a cooling unit configured to cool the preform carried out from the injection molding unit.
6. The apparatus for manufacturing the resin molded article according to claim 5, further comprising: a heating unit configured to continuously convey and heat the preform cooled in the cooling unit; and a blow molding unit configured to blow-mold the preform after heating to manufacture a resin container.
7. The apparatus for manufacturing the resin molded article according to claim 6, wherein a cycle of the blow molding in the blow molding unit is shorter than a cycle of injection molding of the preform in the injection molding unit, and a plurality of the preforms molded in one cycle of the injection molding unit is blow-molded in a plurality of the blow molding cycles.
8. A method for manufacturing a resin molded article using a mold according to claim 1, the method comprising: accommodating and cooling a resin preform having a gate portion protruding outward from a center of a bottom portion in the accommodating space of the mold; sucking air from the air sucking holes of the mold; drawing the preform into the accommodating space; and collapsing the gate portion in the bottom region to remove the gate portion.